4 (a) acceleration/force proportional to distance from a fixed point or displacement M1

either acceleration/force and displacement in opposite directions
or acceleration/force (always) directed towards a fixed point/mean position/equilibrium position A1 [2]

(b) $h\rho g = Mg / A$ B1
$h \times 790 \times 4.9 \times 10^{-4} = 70 \times 10^{-3}$ leading to $h = 0.18\,\text{m}$ or $18\,\text{cm}$ A1 [2]

(c) (i) 1. $\omega^2 = (790 \times 4.9 \times 10^{-4} \times 9.81) / (70 \times 10^{-3})$ C1
$= 54.25$

$\omega = 7.37\ (\text{rad s}^{-1})$
period $(= 2\pi / \omega) = 0.85\,\text{s}$ C1

$t_1 = 0.43$ s A1 [3]

2. $t_3 = 1.28$ s (*allow 2 s.f.*) A1 [1]

(ii) energy of peak $= \frac{1}{2} M\omega^2 x_0^2$ B1

change $= \frac{1}{2} \times 70 \times 10^{-3} \times 54.25\ \{(2.2 \times 10^{-2})^2 - (1.0 \times 10^{-2})^2\}$ C1
$= 7.3 \times 10^{-4}$ J A1 [3]